

Program : Diploma in Polymer Technology	
Course Code : 3097	Course Title: Polymer Analysis Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide skill in the analysis of raw polymer materials.
- To find out the base polymer of the products as to identify an unknown polymer material

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge in chemistry		Applied Chemistry	1
Knowledge in polymer analysis		Fundamentals of polymer science	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify rubbers by chemical analysis methods	26	Applying
CO2	Identify plastics by chemical analysis methods	9	Applying
CO3	Analyses the rubber and plastics products samples	8	Applying
	Lab Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3					
CO3	3	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify rubbers by chemical analysis methods		
MI.01	Identify NBR	3	Applying
M1.02	IdentifyCR	3	Applying
M1.03	IdentifyHypalon(CSM) rubber	3	Applying
M1.04	IdentifySilicone rubber	3	Applying
M1.05	IdentifyEPDM rubber	3	Applying
M1.06	IdentifyBR	3	Applying
M1.07	IdentifyIIR	3	Applying
M1.08	IdentifySBR	3	Applying
M1.09	IdentifyNR/IR	2	Applying
	Lab exam I	1	
CO2	Identify plastics by chemical analysis methods		
M2.01	IdentifyPE	3	Applying
M2.02	IdentifyPET	3	Applying
M2.03	IdentifyNylon	3	Applying
CO3	Analyses the rubber and plastics products samples		
M3.01	Analysis of pressure cooker gasket	3	Applying
M3.02	Analysis of gloves	3	Applying
M3.03	Analysis of plastic products (Bottle,Cover)	2	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	Polymer science V.R.Gowarikar (Publisher – New Age International (P) Ltd, New Delhi - 2005 4
R2	Text Book of Polymer Science -Billmeyer.J.R (John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition
R3	Introductory Polymer Chemistry - G.S.Misra (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition
R4	Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher – Alpha Science International Ltd, 2005 Edition
T1	Polymer science V.R.Gowarikar (Publisher – New Age International (P) Ltd, New Delhi - 2005 4

Online Resources:

Sl.No	Website Link
1	https://link.springer.com
2	https://jordilabs.com